

EE 708: Information Theory and Coding

Midsemester Examination — Model Solutions and Marking Scheme

Spring 2025–26

Question 1**[5 Marks]**

A binary Huffman code must satisfy few conditions: (1) it is *prefix-free* (no codeword is a prefix of another), and from Cover and Thomas Lemma 5.8.1, (2) the two longest codewords have to be of same length and they can only differ in the last bit. Theorem 5.8.1 of Cover and Thomas states that (3) Huffman code is optimal. No coding scheme can achieve lower average length.

(i) {0, 11, 101, 100}

The code is prefix-free. Corresponds to an optimal Huffman code of probability distribution of $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$.

Verdict: YES, this can be a Huffman code.

(ii) {0, 1, 10, 11}

The codeword 1 is a prefix of both 10 and 11. The code is not prefix-free.

Verdict: NO, this cannot be a Huffman code.

(iii) {0, 10, 110, 1110}

The code is prefix-free. However, the two longest codewords are 1110, 110 of different lengths 4 and 3.

Verdict: NO, this cannot be a Huffman code.

(iv) {0, 01, 001, 000}

The codeword 0 is a prefix of 01, 001, and 000. The code is not prefix-free.

Verdict: NO, this cannot be a Huffman code.

(v) {10, 01, 001, 000}

Conditions (1) and (2) satisfy, however (3) fails. The trailing zero at the end of codeword 10 is redundant and could be removed to get lower average length.

Verdict: NO, this cannot be a Huffman code.

Marking Scheme. 1 mark per part for a correct verdict with valid justification. 0.5 mark for correct verdict without valid justification.

Total: 5 marks

Question 2**[3 Marks]**

Claim. For any sequence of sets $B^{(n)} \subseteq \mathcal{X}^n$ with $|B^{(n)}| \leq 2^{nR}$ for all n , if $R < H(X)$, then $\Pr(\mathbf{X} \in B^{(n)}) \not\rightarrow 1$.

Proof.

Suppose, for the sake of contradiction, that there exists a sequence of sets $B^{(n)} \subseteq \mathcal{X}^n$ with $|B^{(n)}| \leq 2^{nR}$ for all n , such that $\Pr(\mathbf{X} \in B^{(n)}) \rightarrow 1$.

For simplicity, assume R is an integer. Label the elements of $B^{(n)}$ as $b(1), b(2), \dots, b(|B^{(n)}|)$ and construct a fixed-to-variable length source coding scheme as follows.

- **Encoder:** Given $\mathbf{x} \in \mathcal{X}^n$, if $\mathbf{x} \in B^{(n)}$, encode it as the index i such that $\mathbf{x} = b(i)$; otherwise, assign the error symbol $\phi(c)$ and declare an error.
- **Decoder:** Given index i , output $b(i)$.

The rate of this scheme satisfies

$$R_n = \frac{1}{n} \log |B^{(n)}| \leq \frac{1}{n} \log 2^{nR} = R.$$

The probability of error satisfies

$$P_e^{(n)} = \Pr(\mathbf{X} \notin B^{(n)}) = 1 - \Pr(\mathbf{X} \in B^{(n)}) \rightarrow 0,$$

by our assumption. Therefore this sequence of codes achieves rate $R_n \leq R < H(X)$ with $P_e^{(n)} \rightarrow 0$. However, this contradicts the **converse of the source coding theorem**, which states that no sequence of codes with rate strictly less than $H(X)$ can achieve $P_e^{(n)} \rightarrow 0$. Hence the assumption is false, and therefore $\Pr(\mathbf{X} \in B^{(n)}) \not\rightarrow 1$. $\square$

Alternate Proof using a Prefix-Free Code Construction.

Suppose, for the sake of contradiction, that $\Pr(\mathbf{X} \in B^{(n)}) \rightarrow 1$, and let $\varepsilon_n = \Pr(\mathbf{X} \notin B^{(n)}) \rightarrow 0$.

Construct a prefix-free code for $\mathbf{X}$ as follows. Label the elements of $B^{(n)}$ as $b(1), b(2), \dots, b(|B^{(n)}|)$.

- For $\mathbf{x} \in B^{(n)}$: assign a prefix-free codeword of length $\lceil nR \rceil + 1$ bits, prepended with the bit “0”, to each $b(i)$, $i = 1, \dots, |B^{(n)}|$. Since $|B^{(n)}| \leq 2^{nR}$, this is feasible.
- For $\mathbf{x} \notin B^{(n)}$: assign a prefix-free codeword of length $\lceil n \log |\mathcal{X}| \rceil + 1$ bits, prepended with the bit “1”, to each remaining sequence. This covers all of $\mathcal{X}^n$ and the two groups are prefix-free from each other by construction.

Since the two groups are distinguished by their leading bit, the overall code is prefix-free and achieves *zero error*. The average codeword length satisfies

$$\bar{\ell}_n = \Pr(\mathbf{X} \in B^{(n)}) \cdot (\lceil nR \rceil + 1) + \Pr(\mathbf{X} \notin B^{(n)}) \cdot (\lceil n \log |\mathcal{X}| \rceil + 1).$$

Using $\Pr(\mathbf{X} \in B^{(n)}) = 1 - \varepsilon_n$ and $\Pr(\mathbf{X} \notin B^{(n)}) = \varepsilon_n$, and bounding $\lceil nR \rceil + 1 \leq nR + 2$ and $\lceil n \log |\mathcal{X}| \rceil + 1 \leq n \log |\mathcal{X}| + 2$:

$$\bar{\ell}_n \leq (1 - \varepsilon_n)(nR + 2) + \varepsilon_n(n \log |\mathcal{X}| + 2) = nR + 2 + \varepsilon_n \cdot n(\log |\mathcal{X}| - R).$$

The average rate of this code is therefore

$$\frac{\bar{\ell}_n}{n} \leq R + \frac{2}{n} + \varepsilon_n \log |\mathcal{X}|.$$

Since $\varepsilon_n \rightarrow 0$ and $\frac{2}{n} \rightarrow 0$, we have

$$\frac{\bar{\ell}_n}{n} \rightarrow R < H(X).$$

However, by the **source coding theorem**, the average rate of any zero-error prefix-free code must satisfy $\frac{\bar{\ell}_n}{n} \geq H(X)$ for all n . This is a contradiction. Hence $\Pr(\mathbf{X} \in B^{(n)}) \not\rightarrow 1$. $\square$

Marking Scheme.

Primary proof — converse of source coding theorem:

- 1 mark — Constructs a valid encoder and decoder using $B^{(n)}$ as the codebook (including handling of sequences outside $B^{(n)}$ via an error symbol), and correctly identifies the rate as $R_n = \frac{1}{n} \log |B^{(n)}| \leq R$.
- 1 mark — Shows $P_e^{(n)} \rightarrow 0$ under the assumption $\Pr(\mathbf{X} \in B^{(n)}) \rightarrow 1$.
- 1 mark — Invokes the converse of the source coding theorem to obtain a contradiction since $R < H(X)$, and concludes $\Pr(\mathbf{X} \in B^{(n)}) \not\rightarrow 1$.

Alternate proof — prefix-free code construction:

- 1 mark — Constructs a valid zero-error prefix-free code using the leading-bit separation trick, correctly assigning codeword lengths $\lceil nR \rceil + 1$ for sequences in $B^{(n)}$ and $\lceil n \log |\mathcal{X}| \rceil + 1$ for sequences outside $B^{(n)}$.
- 1 mark — Correctly computes the average codeword length $\bar{\ell}_n$ and shows that the average rate satisfies $\frac{\bar{\ell}_n}{n} \leq R + \frac{2}{n} + \varepsilon_n \log |\mathcal{X}|$, and concludes that $\frac{\bar{\ell}_n}{n} \rightarrow R < H(X)$.
- 1 mark — Invokes the source coding theorem to obtain a contradiction since any zero-error prefix-free code must achieve average rate $\geq H(X)$, and concludes $\Pr(\mathbf{X} \in B^{(n)}) \not\rightarrow 1$.

Total: 3 marks

General Note on Common Errors (Question 2). The intended proof requires explicitly constructing an encoder–decoder pair using $B^{(n)}$ as the codebook, establishing $P_e^{(n)} \rightarrow 0$ under the assumption $\Pr(\mathbf{X} \in B^{(n)}) \rightarrow 1$, and invoking the *converse of the source coding theorem* to obtain a contradiction. The following frequently occurring approaches are penalised as described.

- **Direct AEP/typical set argument without a coding scheme.**
- **Fano’s inequality without defining an encoder–decoder.** Fano’s inequality requires a specific decoder output $\hat{X}$; invoking it in the abstract is not a valid proof.
- **Treating $B^{(n)}$ as the typical set.** $B^{(n)}$ is an arbitrary subset of $\mathcal{X}^n$ and may contain atypical sequences. Any step relying on this identification is incorrect.
- **Random binning encoder.** Collisions do not vanish when $R < H(X)$, making the error analysis invalid.

Partial credit: For all approaches other than the intended proof, partial credit will be suitably awarded based on the rigour of the reasoning and working presented.

Question 3

[3 Marks]

Let W be an $M \times M$ doubly stochastic matrix and let p, q be probability mass functions on $\{1, \dots, M\}$. Write $(pW)_j = \sum_i p_i W_{ij}$.

Part (a). Show that $D(pW \parallel qW) \leq D(p \parallel q)$.

Apply the log-sum inequality : for non-negative a_i, b_i , $(\sum_i a_i) \log \frac{\sum_i a_i}{\sum_i b_i} \leq \sum_i a_i \log \frac{a_i}{b_i}$. For each fixed j , with $a_i = p_i W_{ij}$ and $b_i = q_i W_{ij}$ we have:

$$(pW)_j \log \frac{(pW)_j}{(qW)_j} \leq \sum_i p_i W_{ij} \log \frac{W_{ij} p_i}{q_i W_{ij}} = \sum_i p_i W_{ij} \log \frac{p_i}{q_i}.$$

Summing over all j and using the row-sum condition $\sum_j W_{ij} = 1$:

$$D(pW \parallel qW) \leq \sum_j \sum_i p_i W_{ij} \log \frac{p_i}{q_i} = \sum_i p_i \log \frac{p_i}{q_i} \underbrace{\left(\sum_j W_{ij} \right)}_{=1} = D(p \parallel q). \square$$

Alternate Solution of part (a) using chain rule on KL Divergence Define joint distributions on $\{1, \dots, M\}^2$ as follows. Let $X \sim p$ and $X' \sim q$, and let $Y \mid X = i$ and $Y' \mid X' = i$ each be distributed according to row i of W , so that

$$p_{XY}(i, j) = p_i W_{ij}, \quad q_{X'Y'}(i, j) = q_i W_{ij}.$$

The marginal distributions of Y and Y' are then pW and qW respectively, since $(pW)_j = \sum_i p_i W_{ij}$ and $(qW)_j = \sum_i q_i W_{ij}$.

Note that the conditional distribution of Y given $X = i$ is identical to that of Y' given $X' = i$: both equal row i of W . That is, $W_{Y|X} = W_{Y'|X'} = W$ (the same stochastic kernel).

Apply the **chain rule for KL divergence** to the joint distributions:

$$D(p_{XY} \parallel q_{X'Y'}) = D(p \parallel q) + D(W_{Y|X} \parallel W_{Y'|X'} \mid p),$$

where the second term is the conditional KL divergence $\mathbb{E}_{X \sim p}[D(W_X \parallel W_X)] = 0$, since the channel W is identical under both joint distributions. Hence

$$D(p_{XY} \parallel q_{X'Y'}) = D(p \parallel q).$$

Applying the chain rule in the other order gives

$$D(p_{XY} \parallel q_{X'Y'}) = D(pW \parallel qW) + D(W_{X|Y} \parallel W_{X'|Y'} \mid pW).$$

Since KL divergence is always non-negative, $D(W_{X|Y} \parallel W_{X'|Y'} \mid pW) \geq 0$. Therefore,

$$D(p \parallel q) = D(pW \parallel qW) + \underbrace{D(W_{X|Y} \parallel W_{X'|Y'} \mid pW)}_{\geq 0} \geq D(pW \parallel qW),$$

which gives $D(pW \parallel qW) \leq D(p \parallel q)$. $\square$

Part (b). Show that $H(pW) \geq H(p)$.

Let $u = (\frac{1}{M}, \dots, \frac{1}{M})$ denote the uniform distribution. Since W is doubly stochastic, its column sums equal 1, so

$$(uW)_j = \sum_i \frac{1}{M} W_{ij} = \frac{1}{M} \sum_i W_{ij} = \frac{1}{M},$$

giving $uW = u$. For any PMF r , the identity $D(r \| u) = \log M - H(r)$ holds. Applying Part (a) with $q = u$:

$$D(pW \| u) = D(pW \| uW) \leq D(p \| u).$$

Substituting the identity:

$$\log M - H(pW) \leq \log M - H(p),$$

which gives $H(pW) \geq H(p)$. □

Marking Scheme.

Part (a) : Log-sum inequality proof:

- **1 mark** : Correctly apply the log-sum inequality to bound each term $(pW)_j \log \frac{(pW)_j}{(qW)_j} \leq \sum_i p_i W_{ij} \log \frac{p_i}{q_i}$.
- **1 mark** : Sum over j and uses the row-sum condition $\sum_j W_{ij} = 1$ to conclude $D(pW \| qW) \leq D(p \| q)$.

Part (a) : Alternate proof via chain rule for KL divergence:

- **1 mark** : Correctly define the joint distributions $p_{XY}(i, j) = p_i W_{ij}$ and $q_{X'Y'}(i, j) = q_i W_{ij}$, identify $Y \sim pW$ and $Y' \sim qW$ as the marginals, and apply the chain rule in the first order to obtain $D(p_{XY} \| q_{X'Y'}) = D(p \| q)$ (using that the conditional KL term vanishes since both joints share the same channel W).
- **1 mark** : Apply the chain rule in the reverse order to write $D(p_{XY} \| q_{X'Y'}) = D(pW \| qW) + D(W_{X|Y} \| W_{X'|Y'} | pW)$, and invoke non-negativity of KL divergence to conclude $D(pW \| qW) \leq D(p \| q)$.

Part (b):

- **1 mark** : Identify $uW = u$, uses $D(r \| u) = \log M - H(r)$, and conclude $H(pW) \geq H(p)$.

Total: 3 marks

Q4:

1 mark for using the general Kraft inequality correctly to get the required inequality.

1 mark for finding the minimum value of D by algebraic methods or by directly substituting in values.

Partial credits: If the binary Kraft inequality is used and some work is shown, **+0.5** is awarded.

Q5:

A complete and proper proof with justifications wherever needed are awarded the full **2 marks**.

Missing justification for a step in the proof gets a **deduction of 0.5**

Incorrect proofs where invalid steps are made to prove the inequality are awarded **0.5 marks** for effort.

Q6:

Drawing the CDF of X correctly gets **1 mark**.

Constructing the SFE codebook:

- **1 mark** for determining the lengths of each codeword properly.
- **2 marks** for determining the midpoint for each X , finding its binary expansion and truncating it correctly to get the codeword.

Cases where errors are made in calculating the length or in calculating the midpoint or in finding the codeword have been awarded partial credits in accordance with the severity of mistakes.

1 mark for correct average length calculation. Mistakes in the codeword length calculation leading to a wrong answer here are awarded **0.5 marks**.

Q4.

TUESDAY

APR '23

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The Kraft inequality for a D-ary code which is uniquely decodable is

$$\sum_i D^{-l_i} \leq 1$$

Codeword lengths are : $\{1, 1, 2, 3, 2, 3\}$

$$\begin{aligned} \therefore D^{-1} + D^{-1} + D^{-2} + D^{-3} + D^{-2} + D^{-3} &\leq 1 \\ \Rightarrow 2(D^{-1} + D^{-2} + D^{-3}) &\leq 1 \end{aligned}$$

Let LHS be given as a function $f(D) = 2(D^{-1} + D^{-2} + D^{-3})$
Clearly, $f(D)$ decreases for increasing $D > 0$.

$$\begin{aligned} \text{Now, } f(2) &= 2(2^{-1} + 2^{-2} + 2^{-3}) = 1.75 > 1 \\ f(3) &= 2(3^{-1} + 3^{-2} + 3^{-3}) = \frac{26}{27} < 1 \end{aligned}$$

$$\begin{aligned} \therefore D_{\min} &= 3 \text{ satisfies Kraft ineq.} \\ \therefore D &\geq 3 \end{aligned}$$

Q5. X is a random variable s.t.

$$P(X = i) = p(i) \quad \text{for } i \in \{1, 2, \dots, m\}$$

$$p_{\max} = \max_i p(i)$$

$$\text{From definition, } H(X) = -\sum_i p(i) \log_2(p(i))$$

$$\begin{aligned} \text{Now, } p(i) &\leq p_{\max} \quad \forall i \in \{1, 2, \dots, m\} \\ \Rightarrow \log_2(p(i)) &\leq \log_2(p_{\max}) \quad (\because \log_2 \text{ is increasing func.}) \end{aligned}$$

$$\Rightarrow \sum_i p(i) \log_2(p(i)) \leq \sum_i p(i) \log_2(p_{\max}) \quad (\because p(i) \geq 0)$$

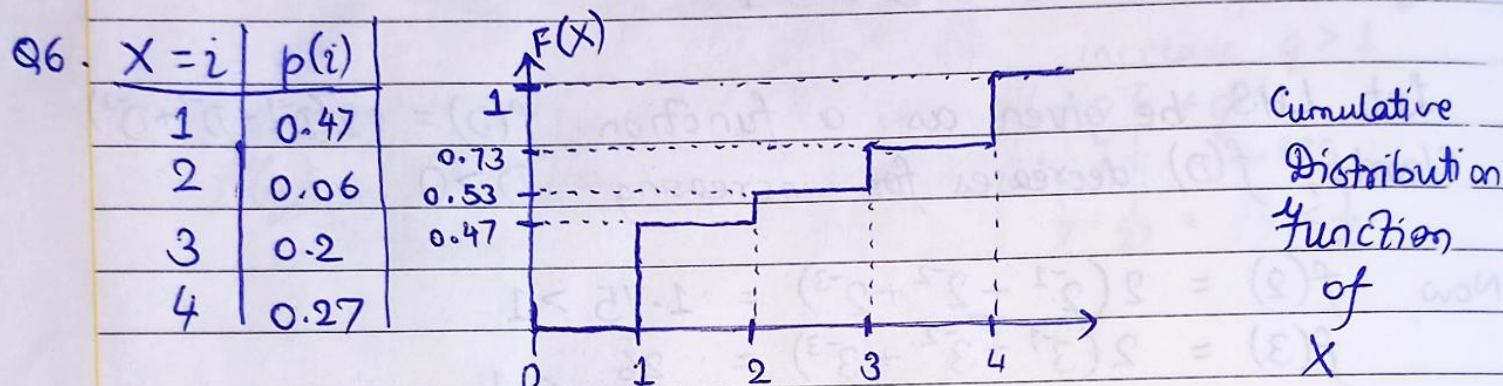
$$\therefore H(X) \geq -\sum_i p(i) \log_2(p_{\max})$$

$$= (-\sum_i p(i)) \log_2(p_{\max})$$

$$= (-1) \cdot \log_2(p_{\max}) \quad (\text{Sum of probability})$$

$$\therefore \boxed{H(X) \geq -\log_2(p_{\max})}$$

Hence shown.



SHANNON - FANO - ELIAS CODE for X.

① Determine the length of each codeword:

$$l(i) = \left\lceil \log_2 \left(\frac{1}{p(i)} \right) \right\rceil + 1 \quad \text{for each } X=i$$

~~∴~~

$$\therefore l(1) = \lceil 1.0893 \rceil + 1 = 3$$

$$l(2) = \lceil 4.059 \rceil + 1 = 6$$

$$l(3) = \lceil 2.322 \rceil + 1 = 4$$

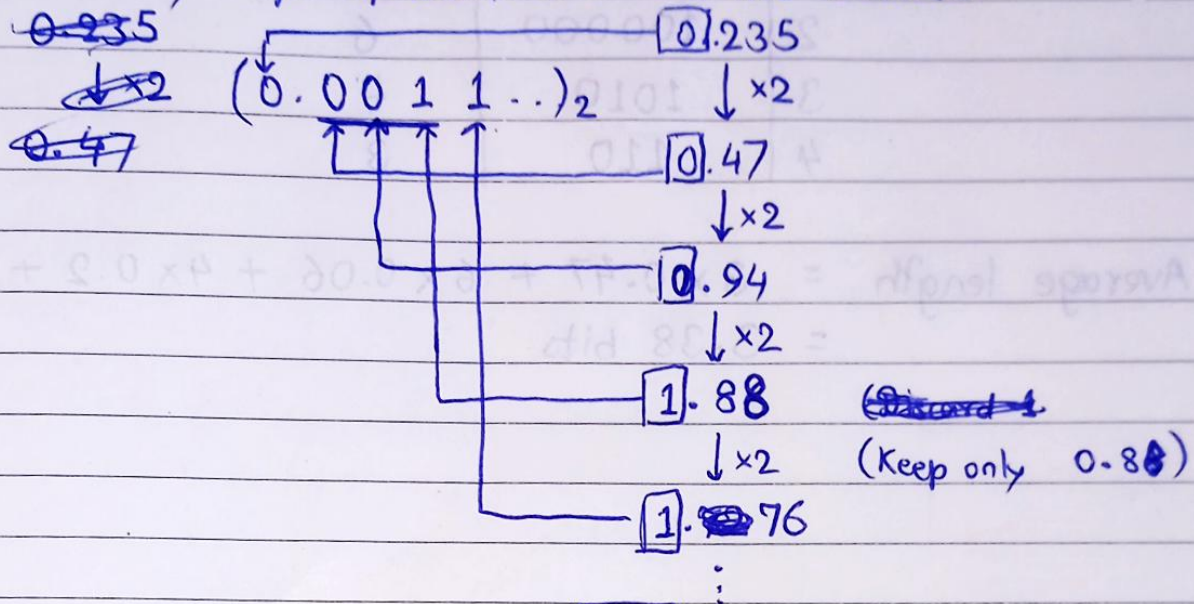
$$l(4) = \lceil 1.889 \rceil + 1 = 3$$

② Find ~~find~~ first $l(i)$ digits of binary expansion of $F(i) - \frac{1}{2} p(i)$ for each $X=i$.

For $X=1$

$$F(1) - \frac{1}{2}p(1) = 0.47 - \frac{0.47}{2} = 0.235$$

To get binary ~~exp~~ representation in the decimals:



Thus codeword for $X=1$ is **001** (3 bits)

$$\text{For } X=2; F(2) - \frac{1}{2}p(2) = 0.53 - \frac{0.06}{2} = 0.5$$

$$(0.5)_{10} \equiv (0.100000...)_2$$

Codeword for $X=2$ is **100000** (6 bits)

$$\text{For } X=3; F(3) - \frac{p(3)}{2} = 0.73 - \frac{0.2}{2} = 0.63$$

$$(0.63)_{10} \equiv (0.1010...)_2$$

Codeword for $X=3$ is **1010** (4 bits)

$$\text{For } X=4; F(4) - \frac{p(4)}{2} = 1 - \frac{0.27}{2} = 0.865$$

$$(0.865)_{10} \equiv (0.110...)_2$$

Codeword for $X=4$ is **110** (3 bits)

FRIDAY

APR '23

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Thus, the SFE Codebook is :

X	Codeword	$l(X=x)$
1	001	3
2	100000	6
3	1010	4
4	110	3

③ Average length = $3 \times 0.47 + 6 \times 0.06 + 4 \times 0.2 + 3 \times 0.27$
= 3.38 bits